Macroeconomics I — MPE Insper

Practice list: Golden rule and the two-period competitive equilibrium (Prova 01, Q2)

Solutions and grading rubric

Instructions. Multiple choice: mark the single correct option (no penalty for wrong answers). *True/False*: every 2 wrong items cancel 1 right one (score = $\max\{0, \text{right} - \text{wrong}/2\}\times$ item value); blank items do not count. Discursive questions: show every step. Parameters are symbols with the stated admissible ranges; answers are closed forms, conditions, signs and limits.

Notation (Kurlat 2020). Capital share α in $Y = AK^{\alpha}L^{1-\alpha}$ (the exam wrote it σ); σ is reserved for the curvature of $u(c) = \frac{c^{1-\sigma}}{1-\sigma}$ ($1/\sigma$ is the intertemporal elasticity of substitution, $\sigma = 1$ is $\ln c$); $\beta \in (0,1)$ discount factor, $\rho \equiv 1/\beta - 1$; δ depreciation; n population growth; a assets carried into the next period; r^K rental rate of capital, r interest rate, with $1 + r = r^K + 1 - \delta$ (Kurlat (9.3.11)). Lower-case k, y, c are per worker unless stated. Book exercises are reproduced verbatim, in the book's own notation, after a source line; parts this list adds are marked.

What this list repairs. (i) Golden-rule algebra: carry the exponent $\frac{\alpha}{1-\alpha}$, cross-check with MPK $= n + \delta$, test a knife edge. (ii) The household problem: every choice variable and every income source, corners included. (iii) Market clearing turns one household's saving rule into a law of motion $k_{t+1} = \phi(k_t)$; the saving rate it implies is chosen, and need not be the golden-rule rate.

Block 1 — Exam level: the golden rule and the two-period household

(Kurlat §§4.3, 6.2; Jones ch. 5 • 35 points)

Question 1. (8 points) *Source: Kurlat, P. (2020),* A Course in Modern Macroeconomics*, Exercise 5.8 "National Accounts and The Golden Rule", pp. 97–98 — statement reproduced verbatim, all items (a)–(d). What this trains:* reading α, s and δ off the national accounts, then deciding on which side of the golden rule the economy sits, by two independent routes (the saving rate against α; the MPK against $n + \delta$).

In 2018, the following events happened in Aurum, which is a closed economy that uses the Denarius (plural: Denarii) as its currency.

- Grano, Inc. hired Cornelia to plant and harvest 500 tons of wheat on a large plot of land using a tractor. Both the land and the tractor are owned by Grano, Inc.
  - It paid Cornelia 200 Denarii for her work on the harvest.
  - It sold the wheat to Panem, Inc. for 1,000 Denarii.
- The tractor the Cornelia used:
  - Was new at the beginning of the year. It was worth 2,000 Denarii.
  - Was no longer new at the end of the year. By then it was worth 1,800 Denarii.
- Panem, Inc. hired Gaius to grind the wheat in its mill and produce a million loaves of bread.
  - It paid Gaius 1,400 Denarii for his work.
  - It sold the bread for 4,000 Denarii to the hungry citizens, who eat it.
- Fabrica, Inc. hired Livia to build a combine harvester.
  - It paid Livia 400 Denarii for her work.
  - It sold the combine harvester to Grano, Inc. for 1,000 Denarii.

(a) Construct GDP accounts (in Denarii) for Aurum by production, income and expenditure.

(b) What were the income shares of labor and capital?

(c) What was the savings rate?

(d) Assume the following:

- the savings rate is constant,
- the production function has the form:

$$Y = AK^{\alpha}L^{1-\alpha}$$

and there is no technological progress,

- the depreciation rate for the tractor is typical for the capital stock as a whole,
- the population is constant.

In the long run, would the level of consumption be higher if the economy slightly increased its saving rate?

Notation.

- Grano, Panem, Fabrica: the three firms of the exercise. VA_j: value added of firm j.
- C, I, G, NX: consumption, investment, government purchases and net exports (final uses); here $G = NX = 0$.
- Y: GDP. K: capital stock. L: labour. $A > 0$: total factor productivity. $\alpha \in (0,1)$: capital share in $Y = AK^{\alpha}L^{1-\alpha}$.
- S: saving. $s \in (0,1)$: saving rate. $\delta \in (0,1)$: depreciation rate. $n \geq 0$: population growth rate.
- k_{SS}, c_{SS}: steady-state capital and consumption per worker. s_{GR}: golden-rule saving rate. MPK: marginal product of capital, $\partial Y/\partial K$.

Step — (a) production. The product approach nets out intermediate inputs:

$$VA_j = \text{sales}_j - \text{intermediate inputs}_j, \qquad \text{GDP} = \textstyle\sum_j VA_j. \tag{1}$$

The tractor is capital; its wear is depreciation, not an intermediate. The wheat is used up within the year, so it is an intermediate input of Panem. Using (1) for each firm:

$$\begin{aligned} VA_{\text{Grano}} &= 1000 - 0 = 1000, \\ VA_{\text{Panem}} &= 4000 - 1000 = 3000, \\ VA_{\text{Fabrica}} &= 1000 - 0 = 1000. \end{aligned}$$

Summing with (1):

$$\text{GDP} = 1000 + 3000 + 1000 = 5000 \text{ Denarii}.$$

Step — (a) income. GDP is also the sum of factor incomes:

$$\text{GDP} = \text{wages} + \text{capital income}. \tag{2}$$

Wages: $200 + 1400 + 400 = 2000$. Capital income is each firm's value added minus its wages (gross operating surplus, depreciation included):

$$\begin{aligned} \text{Grano} &: 1000 - 200 = 800, \\ \text{Panem} &: 3000 - 1400 = 1600, \\ \text{Fabrica} &: 1000 - 400 = 600, \\ \text{total} &: 800 + 1600 + 600 = 3000. \end{aligned}$$

By (2):

$$\text{GDP} = 2000 + 3000 = 5000.$$

Step — (a) expenditure. GDP is the sum of final uses:

$$\text{GDP} = C + I + G + NX. \tag{3}$$

$C = 4000$ (bread eaten by citizens), $I = 1000$ (the combine harvester), $G = NX = 0$. By (3):

$$\text{GDP} = 4000 + 1000 = 5000.$$

The three approaches agree.

Step — (b) shares. Divide each income total of (a) by GDP:

$$\text{labour share} = \frac{2000}{5000} = 0.4, \qquad \text{capital share} = \frac{3000}{5000} = 0.6.$$

With $Y = AK^{\alpha}L^{1-\alpha}$ and competitive factor markets the capital share is α, so $\alpha = 0.6$.

Step — (c) saving rate. Closed economy, no government: saving is output not consumed, and it equals investment,

$$S = Y - C = I. \tag{4}$$

Dividing (4) by Y:

$$s = \frac{I}{Y} = \frac{1000}{5000} = 0.2.$$

Step — (d) route 1, saving rates. The Solow steady state per worker is

$$k_{SS} = \left(\frac{sA}{n+\delta}\right)^{\frac{1}{1-\alpha}}, \tag{5}$$

$$c_{SS} = (1-s)Ak_{SS}^{\alpha}. \tag{6}$$

Substitute (5) into (6); the exponent becomes $\frac{\alpha}{1-\alpha}$:

$$\begin{aligned} c_{SS} &= (1-s)A\left[\left(\frac{sA}{n+\delta}\right)^{\frac{1}{1-\alpha}}\right]^{\alpha} \\ &= (1-s)A\left(\frac{sA}{n+\delta}\right)^{\frac{\alpha}{1-\alpha}} \\ &\propto (1-s)s^{\frac{\alpha}{1-\alpha}}. \end{aligned} \tag{7}$$

With $\alpha = 0.6$ from (b), $\frac{\alpha}{1-\alpha} = 1.5$, so (7) reads

$$\begin{aligned} c_{SS} &\propto (1-s)s^{\frac{(0.6)}{1-(0.6)}} \\ &= (1-s)s^{1.5}. \end{aligned} \tag{8}$$

Differentiate (8) with the product rule:

$$\begin{aligned} \frac{d}{ds}\left[(1-s)s^{1.5}\right] &= 1.5s^{0.5}(1-s) - s^{1.5} \\ &= 1.5s^{0.5} - 2.5s^{1.5} \\ &= s^{0.5}(1.5 - 2.5s). \end{aligned} \tag{9}$$

(9) is zero at $s_{GR} = 1.5/2.5 = 0.6 = \alpha$ and positive for $s < 0.6$. At $s = 0.2$:

$$\begin{aligned} (0.2)^{0.5}(1.5 - 2.5(0.2)) &= 0.2^{0.5}(1.5 - 0.5) \\ &= 0.447 > 0. \end{aligned}$$

Step — (d) route 2, MPK. The tractor, worth 2000 when new, is worth 1800 at year end, and the population is constant:

$$\delta = \frac{2000 - 1800}{2000} = 0.1, \qquad n = 0.$$

In the steady state with $n = 0$, saving replaces depreciation:

$$sY = \delta K. \tag{10}$$

Divide (10) by δY:

$$\frac{K}{Y} = \frac{s}{\delta} = \frac{0.2}{0.1} = 2.$$

Differentiating $Y = AK^{\alpha}L^{1-\alpha}$ with respect to K gives

$$\text{MPK} = \alpha\frac{Y}{K}. \tag{11}$$

Substituting $\alpha = 0.6$ and $K/Y = 2$ into (11):

$$\begin{aligned} \text{MPK} &= \frac{(0.6)}{(2)} \\ &= 0.3 > n + \delta = 0.1. \end{aligned}$$

Too little capital: setting (11) equal to $n + \delta = \delta$, the golden rule needs

$$\begin{aligned} (0.6)\frac{Y}{K} &= (0.1) \\ \frac{K}{Y} &= \frac{\alpha}{\delta} = 6. \end{aligned}$$

Step — answer. Yes: a slightly higher saving rate raises long-run consumption, because $s = 0.2 < s_{GR} = \alpha = 0.6$ by (9) (equivalently MPK $= 0.3 > n + \delta = 0.1$ by (11)).

Economic intuition. Below the golden rule an extra unit of capital yields more output (MPK) than it costs to maintain $(n + \delta)$, so the steady state can afford both more capital and more consumption.

Ref: Kurlat (2020), §4.3, eq. (4.3.4), p. 61. Matches Problem 5.8 in the project's worked solutions, `Resolucao/kurlat_solutions_ch05.tex`.

Grading rubric.

- (a) three approaches, each computed: 3 pts. Wheat treated as final or the harvester netted out: max. 1 pt for (a).
- (b)–(c): 2 pts. (d) both routes: 3 pts; verdict without a computed comparison: max. 1 pt for (d).

Question 2. (10 points) *Source: Jones, C. I. (2020),* Macroeconomics*, 5th ed., ch. 5, Review Question 4, p. 129 — statement reproduced verbatim; item (a) asks it in Jones's model with the exponent* 1/3*; items (b)–(d) are added by this list. What this trains:* the steady-state consumption formula with the right exponent, and the exam's exact algebra slip.

Review Question 4 (Jones). What is the solution of the Solow model for consumption per person in the steady state, $c^* \equiv C^*/L^*$? How does each parameter in the solution affect c^*, and why?

a) (2 points) Answer Review Question 4 in Jones's model ($Y = \bar{A}K^{1/3}\bar{L}^{2/3}$, $\Delta K = sY - \bar{d}K$).

Added parts (not in the book):

b) (2 points) Find the s that maximises c^*. Check it with the MPK condition.

c) (3 points) In the exam, a student substituted k^* instead of $k^{*\alpha}$ and maximised $(1-s)\left(\frac{s}{n+\delta}\right)^{\frac{1}{1-\alpha}}$. For a general capital share α, which saving rate does that give? Compare it with the true golden rule as $\alpha \to 0$ and explain which one passes this knife-edge test, and why.

d) (3 points) Let $Y_t = K_t^{\alpha}(A_t L_t)^{1-\alpha}$, $A_{t+1} = (1+g)A_t$, $L_{t+1} = (1+n)L_t$, $K_{t+1} = (1-\delta)K_t + sY_t$. With $\tilde{k} \equiv K/(AL)$, derive $\tilde{k}_{t+1}(\tilde{k}_t)$, the steady state $\tilde{k}^*$, consumption per effective worker $\hat{c}^*(s)$ and the golden-rule s. Show that at any steady state MPK $= \alpha N/s$, with $N \equiv (1+n)(1+g) - 1 + \delta$. Evaluate $\hat{c}^*$ at $s \in \{0.2,\ s_{GR},\ 0.6\}$ for $\alpha = 1/3$, $n = 0.01$, $g = 0.02$, $\delta = 0.05$.

Notation.

- Y: output. K: capital. $\bar{L} > 0$: labour, constant. $\bar{A} > 0$: TFP, constant. ΔK: change in K over a period.
- $s \in (0,1)$: saving rate. $\bar{d} \in (0,1)$: Jones's depreciation rate (Kurlat's δ).
- K^*: steady-state capital; $y^* \equiv Y^*/\bar{L}$, c^*: steady-state output and consumption per worker.
- Items (c)–(d): $\alpha \in (0,1)$ capital share; $n > 0$, $g > 0$: growth rates of L and A; $\delta \in (0,1)$ depreciation.
- $\tilde{k} \equiv K/(AL)$, $\tilde{y} \equiv Y/(AL)$: capital and output per effective worker; $\hat{c}^*$: steady-state consumption per effective worker. $N \equiv (1+n)(1+g) - 1 + \delta$: the effective break-even rate. MPK $= \partial Y/\partial K$.

Step — (a) steady state. Jones's model, with $\bar{L}$ constant:

$$Y = \bar{A}K^{1/3}\bar{L}^{2/3}, \tag{12}$$

$$\Delta K = sY - \bar{d}K. \tag{13}$$

Set $\Delta K = 0$ in (13):

$$sY^* = \bar{d}K^*. \tag{14}$$

Substitute (12) into (14):

$$s\big(\bar{A}K^{*1/3}\bar{L}^{2/3}\big) = \bar{d}K^*.$$

Divide by $\bar{d}K^{*1/3}$:

$$K^{*2/3} = \frac{s\bar{A}}{\bar{d}}\,\bar{L}^{2/3}.$$

Raise both sides to the power $3/2$ and divide by $\bar{L}$:

$$\frac{K^*}{\bar{L}} = \left(\frac{s\bar{A}}{\bar{d}}\right)^{3/2}. \tag{15}$$

Divide (12) by $\bar{L}$ and substitute (15):

$$\begin{aligned} y^* &= \bar{A}\left(\frac{K^*}{\bar{L}}\right)^{1/3} \\ &= \bar{A}\left[\left(\frac{s\bar{A}}{\bar{d}}\right)^{3/2}\right]^{1/3} \\ &= \bar{A}\left(\frac{s\bar{A}}{\bar{d}}\right)^{1/2} \\ &= \bar{A}^{3/2}\left(\frac{s}{\bar{d}}\right)^{1/2}. \end{aligned} \tag{16}$$

Consumption is the share $1 - s$ of (16):

$$\begin{aligned} c^* &= (1-s)y^* \\ &= (1-s)\left[\bar{A}^{3/2}\left(\frac{s}{\bar{d}}\right)^{1/2}\right] \\ &= (1-s)\bar{A}^{3/2}\left(\frac{s}{\bar{d}}\right)^{1/2}. \end{aligned} \tag{17}$$

Step — (a) parameters. Read the exponents of (17). $\partial c^*/\partial \bar{A} > 0$ with elasticity $3/2$: TFP raises output directly and through the capital it induces. $\partial c^*/\partial \bar{d} < 0$ (elasticity $-1/2$): more depreciation means less capital. s: ambiguous, a larger capital stock against a smaller consumed share.

Step — (b) maximise. Take the log of (17):

$$\ln c^* = \text{const} + \ln(1-s) + \tfrac{1}{2}\ln s. \tag{18}$$

Differentiate (18) with respect to s and set it to zero:

$$-\frac{1}{1-s} + \frac{1}{2s} = 0. \tag{19}$$

Multiply (19) by $2s(1-s) > 0$:

$$\begin{aligned} -2s + (1-s) &= 0 \\ 2s &= 1 - s \\ s &= \tfrac{1}{3}. \end{aligned}$$

SOC: differentiating (19) once more, $-\frac{1}{(1-s)^2} - \frac{1}{2s^2} < 0$, a maximum. Check with the MPK. Differentiating (12) and using (14):

$$\begin{aligned} \text{MPK} &= \frac{1}{3}\frac{Y}{K} \\ &= \frac{1}{3}\left(\frac{\bar{d}}{s}\right). \end{aligned} \tag{20}$$

Substituting $s = 1/3$ into (20):

$$\begin{aligned} \text{MPK} &= \frac{1}{3}\left(\frac{\bar{d}}{(1/3)}\right) \\ &= \bar{d}, \end{aligned}$$

so MPK equals $\bar{d}$ (here $n = 0$) exactly at $s = 1/3$.

Step — (c) the slip. The student's objective, in logs:

$$\ln\left[(1-s)s^{\frac{1}{1-\alpha}}\right] = \ln(1-s) + \frac{1}{1-\alpha}\ln s. \tag{21}$$

Differentiate (21) and set it to zero:

$$-\frac{1}{1-s} + \frac{1}{(1-\alpha)s} = 0.$$

Multiply by $(1-\alpha)s(1-s) > 0$:

$$\begin{aligned} (1-\alpha)s &= 1-s \\ s &= \frac{1}{2-\alpha}. \end{aligned} \tag{22}$$

The right objective carries the exponent $\frac{\alpha}{1-\alpha}$:

$$\ln\left[(1-s)s^{\frac{\alpha}{1-\alpha}}\right] = \ln(1-s) + \frac{\alpha}{1-\alpha}\ln s. \tag{23}$$

Differentiate (23), set it to zero and multiply by $(1-\alpha)s(1-s)$:

$$\begin{aligned} \alpha(1-s) &= (1-\alpha)s \\ s_{GR} &= \alpha. \end{aligned} \tag{24}$$

At $\alpha = 1/3$, (22) gives

$$\begin{aligned} s &= \frac{1}{2-(1/3)} \\ &= 0.6, \end{aligned}$$

against $s_{GR} = 1/3$ from (24).

Step — (c) knife edge. As $\alpha \to 0$ capital produces nothing, so every unit saved is wasted and the best s must go to 0: (24) gives $s_{GR} = \alpha \to 0$ and passes; (22) gives $\frac{1}{2-\alpha} \to \frac{1}{2}$ and fails (it would save half of output to maintain useless machines).

Step — (d) law of motion. The statement's model:

$$Y_t = K_t^{\alpha}(A_t L_t)^{1-\alpha}, \tag{25}$$

$$K_{t+1} = (1-\delta)K_t + sY_t, \tag{26}$$

$$A_{t+1}L_{t+1} = (1+g)(1+n)A_t L_t. \tag{27}$$

Divide (25) by A_tL_t:

$$\tilde{y}_t = \tilde{k}_t^{\alpha}. \tag{28}$$

Divide (26) by $A_{t+1}L_{t+1}$, written as in (27), and use (28):

$$\begin{aligned}
\tilde{k}_{t+1} &= \frac{(1-\delta)K_t + sY_t}{((1+g)(1+n)A_tL_t)} \\
&= \frac{(1-\delta)\tilde{k}_t + s\tilde{y}_t}{(1+n)(1+g)} \\
&= \frac{(1-\delta)\tilde{k}_t + s(\tilde{k}_t^{\alpha})}{(1+n)(1+g)} \\
&= \frac{(1-\delta)\tilde{k}_t + s\tilde{k}_t^{\alpha}}{(1+n)(1+g)}.
\end{aligned} \tag{29}$$

Steady state: set $\tilde{k}_{t+1} = \tilde{k}_t = \tilde{k}$ in (29) and multiply by $(1+n)(1+g)$:

$$\begin{aligned}
(\tilde{k}) &= \frac{(1-\delta)(\tilde{k}) + s(\tilde{k})^{\alpha}}{(1+n)(1+g)} \\
(1+n)(1+g)\tilde{k} &= (1-\delta)\tilde{k} + s\tilde{k}^{\alpha} \\
\big[(1+n)(1+g) - 1 + \delta\big]\tilde{k} &= s\tilde{k}^{\alpha} \\
N\tilde{k} &= s\tilde{k}^{\alpha}.
\end{aligned} \tag{30}$$

Solve (30) for $\tilde{k}$:

$$\tilde{k}^* = \left(\frac{s}{N}\right)^{\frac{1}{1-\alpha}}. \tag{31}$$

Consumption per effective worker is $(1-s)\tilde{y}$; use (28) and (31):

$$\begin{aligned}
\hat{c}^* &= (1-s)(\tilde{k}^{*\alpha}) \\
&= (1-s)\left[\left(\frac{s}{N}\right)^{\frac{1}{1-\alpha}}\right]^{\alpha} \\
&= (1-s)\left(\frac{s}{N}\right)^{\frac{\alpha}{1-\alpha}}.
\end{aligned} \tag{32}$$

In s, (32) is the objective (23) up to a constant, so it is maximised at $s_{GR} = \alpha$, (24). Expanding, $N = n + g + ng + \delta$: the continuous-time $n + g + \delta$ plus the cross term ng.

Step — (d) MPK cross-check. Differentiate (25) with respect to K_t:

$$\text{MPK} = \alpha K^{\alpha-1}(AL)^{1-\alpha} = \alpha\tilde{k}^{\alpha-1}. \tag{33}$$

By (31), $\tilde{k}^{*1-\alpha} = s/N$. Substitute into (33):

$$\begin{aligned}
\text{MPK} &= \frac{\alpha}{\tilde{k}^{*1-\alpha}} \\
&= \frac{\alpha}{(s/N)} \\
&= \frac{\alpha N}{s}.
\end{aligned} \tag{34}$$

So $\text{MPK} = N \iff s = \alpha$; $s < \alpha \iff \text{MPK} > N$ (too little capital).

Step — (d) numbers. The break-even rate:

$$N = 1.01 \times 1.02 - 1 + 0.05 = 0.0802.$$

With $\alpha = 1/3$, $\frac{1}{1-\alpha} = 1.5$. Using (31):

$$s = 0.2: \ \tilde{k}^* = (0.2/0.0802)^{1.5} = 3.9381,$$
$$s = 1/3: \ \tilde{k}^* = (0.3333/0.0802)^{1.5} = 8.4734,$$
$$s = 0.6: \ \tilde{k}^* = (0.6/0.0802)^{1.5} = 20.4628.$$

Using (32) as $(1-s)\tilde{k}^{*1/3}$:

$$s = 0.2: \ \hat{c}^* = 0.8 \times 3.9381^{1/3} = 1.2633,$$
$$s = 1/3: \ \hat{c}^* = \tfrac{2}{3} \times 8.4734^{1/3} = 1.3591,$$
$$s = 0.6: \ \hat{c}^* = 0.4 \times 20.4628^{1/3} = 1.0941.$$

Using (34):

$$s = 0.2: \ \text{MPK} = \frac{(1/3)(0.0802)}{(0.2)} = 0.1337,$$
$$s = 1/3: \ \text{MPK} = \frac{(1/3)(0.0802)}{(1/3)} = 0.0802 = N,$$
$$s = 0.6: \ \text{MPK} = \frac{(1/3)(0.0802)}{(0.6)} = 0.0446.$$

The slip's answer ($s = 0.6$, from (22)) loses

$$1 - \frac{1.0941}{1.3591} = 19.5\%$$

of the attainable consumption.

Economic intuition. The golden rule reinvests exactly capital's income: saving α of output pays the maintenance $N\tilde{k}$ of the stock whose marginal product equals that maintenance cost.

Ref: Jones (2020), ch. 5, eqs. (5.7)–(5.9), pp. 112–113; Jones, *Instructor's Manual* (2016), ch. 5, answer to Review Question 4, p. 37 (gives $c^* = (1-s)\bar{A}^{3/2}(s/\bar{d})^{1/2}$ and "optimal s = the exponent on capital, 1/3"; its line that a higher depreciation rate *raises* c^* is a slip: $\bar{d}$ is in the denominator). Kurlat (2020), §4.3; Romer (2012), Problem 2.15(c), p. 97.

Grading rubric.

- (a) c^* with exponents 3/2, 1/2: 2 pts. (b): 2 pts; no MPK check: −0.5.
- (c) $1/(2-\alpha)$ and the knife-edge argument: 3 pts. (d): 3 pts; arithmetic slip with correct method: −0.5 (does not add to a ceiling).
- Ceiling: maximising $(1-s)k^*$ instead of $(1-s)k^{*\alpha}$ anywhere: max. 1 pt on that sub-item.

Question 3. (10 points) *Source: Kurlat, P. (2020),* A Course in Modern Macroeconomics*, Exercise 6.1 "Two-Period Problem with Taxes and Initial Wealth", pp. 121–122 — statement reproduced verbatim, all items (a)–(e). What this trains:* the full household problem with every income source, closed forms, and how σ decides whether c_1 rises or falls with r (the log case $\sigma = 1$ is the exam's). Suppose a household solves the following two-period consumption-savings problem with taxes:

$$\max_{c_1, a, c_2} \ u(c_1) + \beta u(c_2)$$
$$\text{s.t.}$$
$$a = a_0 + y_1 - \tau_1 - c_1$$
$$c_2 = y_2 - \tau_2 + (1+r)\, a$$

with $u(c) = \frac{c^{1-\sigma}}{1-\sigma}$, where: c_1 is consumption at time 1, c_2 is consumption at time 2, y_1 is household income at time 1, y_2 is household income at time 2, τ_1 are taxes at time 1, τ_2 are taxes at time 2, and a_0 is initial wealth.

(a) Solve for the household's choice of c_1, c_2 and a in closed form.[1]

(b) How does $\frac{c_1}{y_1}$ depend on y_2? What would happen if households suddenly became optimistic about the future?

(c) How does $\frac{c_1}{y_1}$ depend on a_0? Interpret.

(d) How does $\frac{c_1}{y_1}$ depend on β? Interpret.

(e) Suppose $y_2 = \tau_1 = \tau_2 = 0$ and compute $\frac{\partial c_1}{\partial r}$. How does the answer depend on σ? Interpret the answer. [Hint: this is a hard question, not the maths but the interpretation. Think about what σ means for the relative importance of income and substitution effects]

Notation.

- $c_1, c_2 > 0$: consumption in periods 1 and 2. a: assets carried into period 2 ($a < 0$ is borrowing).
- $a_0 \geq 0$: initial wealth. $y_1, y_2 \geq 0$: income. τ_1, τ_2: lump-sum taxes. $r > -1$: interest rate.
- $\beta \in (0,1)$: discount factor. $u(c) = \frac{c^{1-\sigma}}{1-\sigma}$, $\sigma > 0$ (curvature; $1/\sigma$ is the intertemporal elasticity of substitution; $\sigma = 1$ is $\ln c$), so $u'(c) = c^{-\sigma}$.
- W: present value of resources; D: the consumption divisor (both defined below, where first used).

Step — (a) set-up. Choice variables c_1, c_2, a. The two period budgets:

$$c_1 + a = a_0 + y_1 - \tau_1, \tag{35}$$

$$c_2 = y_2 - \tau_2 + (1+r)a. \tag{36}$$

Solve (35) for a and substitute into (36):

$$c_2 = y_2 - \tau_2 + (1+r)(a_0 + y_1 - \tau_1 - c_1). \tag{37}$$

With c_2 given by (37), the problem has one choice:

$$\max_{c_1} \; u(c_1) + \beta u(c_2), \qquad c_2 \text{ from (37)}. \tag{38}$$

Step — (a) Euler. Differentiate (38) with respect to c_1, using $\partial c_2/\partial c_1 = -(1+r)$ from (37):

$$u'(c_1) - \beta(1+r)u'(c_2) = 0. \tag{39}$$

Substitute $u'(c) = c^{-\sigma}$ into (39):

$$\begin{aligned} (c_1^{-\sigma}) - \beta(1+r)(c_2^{-\sigma}) &= 0 \\ c_1^{-\sigma} &= \beta(1+r)c_2^{-\sigma}. \end{aligned}$$

Divide by $c_1^{-\sigma}\beta(1+r)$ and raise to the power $1/\sigma$:

$$c_2 = [\beta(1+r)]^{1/\sigma} c_1. \tag{40}$$

Step — (a) closed forms. Divide (37) by $1+r$ and move c_1 to the left: the present-value budget is

$$c_1 + \frac{c_2}{1+r} = W, \qquad W \equiv a_0 + y_1 - \tau_1 + \frac{y_2 - \tau_2}{1+r}. \tag{41}$$

Substitute (40) into (41):

$$c_1 + \frac{([\beta(1+r)]^{1/\sigma} c_1)}{1+r} = W$$

[1] Kurlat's footnote: "In closed form" means that you have an explicit expression for something. Suppose I ask you to determine some variable x. Then something like $x = y^2 - z$ is a solution "in closed form"; something like $e^x + x - b = 0$ also implicitly tells what x should be but is not in closed form.

$$c_1 + \frac{\beta^{1/\sigma}(1+r)^{1/\sigma}}{1+r}\, c_1 = W$$
$$c_1\big[1 + \beta^{1/\sigma}(1+r)^{\frac{1}{\sigma}-1}\big] = W.$$

Name the bracket

$$D \equiv 1 + \beta^{1/\sigma}(1+r)^{\frac{1}{\sigma}-1} > 1. \tag{42}$$

Then

$$c_1 = \frac{W}{D}. \tag{43}$$

Substitute (43) into (40), and into (35) solved for a:

$$\begin{aligned} c_2 &= [\beta(1+r)]^{1/\sigma}\left(\frac{W}{D}\right) \\ &= \frac{[\beta(1+r)]^{1/\sigma}W}{D}, \\ a &= a_0 + y_1 - \tau_1 - \left(\frac{W}{D}\right) \\ &= a_0 + y_1 - \tau_1 - \frac{W}{D}. \end{aligned}$$

Step — (b) future income. By (41), $\partial W/\partial y_2 = \frac{1}{1+r}$. Differentiate (43):

$$\begin{aligned} \frac{\partial c_1}{\partial y_2} &= \frac{1}{D}\left(\frac{1}{1+r}\right) \\ &= \frac{1}{(1+r)D} > 0, \\ \frac{\partial (c_1/y_1)}{\partial y_2} &= \frac{1}{y_1(1+r)D}. \end{aligned}$$

Optimism about y_2 raises consumption and cuts saving today.

Step — (c) wealth. By (41), $\partial W/\partial a_0 = \partial W/\partial y_1 = 1$. Differentiate (43):

$$\begin{aligned} \frac{\partial c_1}{\partial a_0} = \frac{\partial c_1}{\partial y_1} &= \frac{1}{D}\,(1) \\ &= \frac{1}{D} \in (0,1), \end{aligned}$$

using $D > 1$ from (42). Only the present value of resources matters, and the household spreads any windfall over both periods.

Step — (d) patience. Differentiate (42) with respect to β:

$$\frac{\partial D}{\partial \beta} = \frac{1}{\sigma}\beta^{\frac{1}{\sigma}-1}(1+r)^{\frac{1}{\sigma}-1} > 0.$$

Differentiate (43) (W does not contain β) and substitute:

$$\frac{\partial c_1}{\partial \beta} = -\frac{W}{D^2}\cdot\frac{\partial D}{\partial \beta} = -\frac{W}{D^2}\cdot\left(\frac{1}{\sigma}\beta^{\frac{1}{\sigma}-1}(1+r)^{\frac{1}{\sigma}-1}\right) < 0.$$

A more patient household consumes less now and saves more.

Step — (e) interest rate. With $y_2 = \tau_1 = \tau_2 = 0$, (41) gives

$$\begin{aligned} W &= a_0 + y_1 - (0) + \frac{(0)-(0)}{1+r} \\ &= a_0 + y_1, \end{aligned}$$

so (43) becomes

$$c_1 = \frac{(a_0 + y_1)}{D}. \tag{44}$$

Differentiate (42) with respect to r:

$$D'(r) = \beta^{1/\sigma}\left(\frac{1}{\sigma} - 1\right)(1+r)^{\frac{1}{\sigma}-2}. \tag{45}$$

Differentiate (44) and substitute (45), using $-\left(\frac{1}{\sigma} - 1\right) = \frac{\sigma-1}{\sigma}$:

$$\begin{aligned}\frac{\partial c_1}{\partial r} &= -\frac{(a_0 + y_1)D'(r)}{D^2} \\ &= -\frac{(a_0 + y_1)\left(\beta^{1/\sigma}\left(\frac{1}{\sigma} - 1\right)(1+r)^{\frac{1}{\sigma}-2}\right)}{D^2} \\ &= \frac{(a_0 + y_1)\beta^{1/\sigma}(1+r)^{\frac{1}{\sigma}-2}}{D^2} \cdot \frac{\sigma - 1}{\sigma}.\end{aligned} \tag{46}$$

Sign of (46) = sign of $\sigma - 1$. A saver is richer when r rises (income effect: $c_1 \uparrow$), and waiting is cheaper (substitution: $c_1 \downarrow$). $\sigma > 1$ (low elasticity $1/\sigma$): income wins. $\sigma < 1$: substitution wins. $\sigma = 1$: they cancel; (42) gives

$$\begin{aligned}D &= 1 + \beta^{1/(1)}(1+r)^{\frac{1}{(1)}-1} \\ &= 1 + \beta,\end{aligned}$$

and $c_1 = \frac{a_0+y_1}{1+\beta}$ does not depend on r.

Economic intuition. Numerator W = how much the household has; denominator D = how it splits it across dates. Every comparative static moves one of the two.

Ref: Kurlat (2020), §6.2, pp. 105–117. Instructor key PSET 3 (`Listas/soluções do instrutor/PSET 3 (1).pdf`), Q1(a)–(c): same method, same c_1 and a. Two slips in that key: its boxed c_2 has denominator $1 + \beta^{1/\sigma}(1+r)^{\frac{1}{\sigma}-1}$ where $1 + \beta^{-1/\sigma}(1+r)^{1-\frac{1}{\sigma}}$ is needed, and its $\partial c_1/\partial r$ drops the factor $\beta^{1/\sigma}$; the sign, set by $\frac{\sigma-1}{\sigma}$, is right.

Grading rubric.

- (a) Euler and closed forms: 4 pts. Euler only, no closed forms: max. 2 pts.
- (b)–(d): 3 pts. (e) derivative and its sign: 2 pts; income and substitution reading: 1 pt.
- Ceiling: $1/\sigma$ and σ swapped in the reading of (e): max. 1 pt for (e).

Question 4. (7 points) *Source: Kurlat, P. (2020),* A Course in Modern Macroeconomics*, Exercise 6.2 "Savings Rates for Different People", p. 122 — statement reproduced verbatim, all items (a)–(b). Its Exercise 6.1 is Question 3 of this list. What this trains:* a saving rate out of *current* income depends on the income profile and, through human wealth, on r, even when the saving rate out of *lifetime* wealth does not.

An economist has data on the occupation, this year's income (denoted y) and this year's consumption (denoted c) of a sample of 26-year-olds. Within this sample, some are top professional athletes and others are medical doctors in their first year of residence. [Make whatever assumptions you think are reasonable and, if you wish, refer back to Exercise 6.1]

(a) Suppose one computed $s = \frac{y-c}{y}$ for each individual in the sample. Should we expect s to be higher for athletes or doctors?

(b) Suppose interest rates go down. How should we expect the response of s to differ between the two groups?

Notation.

- $y_1, y_2 > 0$: income in periods 1 and 2; $m \equiv y_2/y_1$: the income profile. c_1: consumption in period 1.
- $s \equiv 1 - c_1/y_1$: saving rate out of *current* income ($s < 0$ is borrowing).

- $r > -1$, $\beta \in (0,1)$, $\sigma > 0$, W and D: as in Question 3, (41) and (42).

Step — (a) assumption. A 26-year-old athlete earns most of his lifetime income now (m small); a first-year resident earns little now and much later (m large). Use Question 3 with $a_0 = \tau_1 = \tau_2 = 0$; then (41) gives

$$W = y_1 + \frac{y_2}{1+r} = y_1\Big(1 + \frac{m}{1+r}\Big). \tag{47}$$

The saving rate out of current income is

$$s = 1 - \frac{c_1}{y_1}. \tag{48}$$

Substitute (43) and then (47) into (48):

$$\begin{aligned} s &= 1 - \frac{1}{y_1}\left(\frac{W}{D}\right) \\ &= 1 - \frac{1}{y_1 D}\left(y_1\Big(1 + \frac{m}{1+r}\Big)\right) \\ &= 1 - \frac{1}{D}\Big(1 + \frac{m}{1+r}\Big), \end{aligned} \tag{49}$$

which is decreasing in m.

Step — (a) answer. s is higher for athletes; residents' s can be negative (they borrow). With $\beta = 0.97$, $\sigma = 2$, $r = 0.06$, (42) gives $D = 1 + 0.97^{1/2}\,1.06^{-1/2} = 1.9566$. Using (49):

$$\text{athlete } (y_1, y_2) = (10, 2),\ m = 0.2:\ \ s = 1 - \frac{1 + 0.2/1.06}{1.9566} = 0.392,$$
$$\text{resident } (3, 9),\ m = 3:\ \ s = 1 - \frac{1 + 3/1.06}{1.9566} = -0.958.$$

Step — (b) channels. Differentiate (47) with respect to r:

$$\frac{\partial W}{\partial r} = -\frac{y_2}{(1+r)^2} < 0.$$

So a fall in r (i) raises human wealth $\frac{y_2}{1+r}$, by an amount proportional to y_2, raising c_1; (ii) changes D, whose sign depends on σ (Question 3(e), (45)). For borrowers (residents) income and substitution effects both raise c_1; for lenders (athletes) they conflict.

Step — (b) answer. Residents' s falls clearly; athletes' s barely moves (and, with $\sigma > 1$, rises). Cut r from 6% to 2%; (42) and then (49) give:

$$D = 1 + 0.97^{1/2}\,(1.02)^{-1/2} = 1.9752,$$
$$\text{athlete}:\ \ s = 1 - \frac{1 + 0.2/(1.02)}{(1.9752)} = 0.3944,$$
$$\text{resident}:\ \ s = 1 - \frac{1 + 3/(1.02)}{(1.9752)} = -0.9953.$$

Subtracting the values of (a) to four decimals (0.3925 and −0.9576):

$$\text{athlete } \Delta s = +0.002, \qquad \text{resident } \Delta s = -0.038.$$

Economic intuition. The same optimising rule produces opposite saving rates when income arrives at different dates; a rate cut is a transfer from lenders to borrowers.

Ref: Kurlat (2020), §6.2; numbers as in the project's worked solution `Resolucao/kurlat_solutions_ch06.tex`, Problem 6.2.

Grading rubric.

- (a) the income-profile argument with the formula for s: 3 pts; ranking without the profile argument: max. 1 pt.
- (b) human-wealth channel and borrower/lender sign: 4 pts; "lower r always lowers saving" for both groups: max. 1 pt.

Block 2 — Harder: equilibrium, endogenous saving, and the golden rule

(Kurlat §§9.1–9.3; Romer §2.9–2.11 • 37 points)

Question 5. (6 points) *Source: Kurlat, P. (2020),* A Course in Modern Macroeconomics*, Exercise 9.2 "Storage", pp. 179–180 — statement reproduced verbatim, all items (a)–(c). What this trains:* prices come from the firm's FOCs and the no-arbitrage condition $1 + r = r^K + 1 - \delta$; the Euler equation then signs the consumption path.
Suppose the production function is the following:

$$F(K, L) = K$$

and the rate of depreciation is $\delta = 1$.

(a) Why is this production function called "storage"?

(b) What will be the real interest rate and the wage in this economy?

(c) With standard preferences, will consumption increase over time, remain constant over time or decrease over time?

Notation.

- $F(K, L)$: production function; F_K, F_L: marginal products of capital and labour. $\delta \in [0, 1]$: depreciation rate.
- r^K: rental rate of capital. w: wage. r: interest rate.
- $c_t > 0$: consumption at t. u: utility, $u' > 0$, $u'' < 0$. $\beta \in (0, 1)$: discount factor; $\rho \equiv 1/\beta - 1 > 0$.

Step — (a). One unit of goods put aside as capital today returns exactly one unit tomorrow and then disappears; labour is useless:

$$F_K = 1, \qquad F_L = 0, \qquad \delta = 1. \tag{50}$$

The technology only moves goods through time.

Step — (b). The firm's FOCs set each factor price equal to its marginal product:

$$r^K = F_K, \qquad w = F_L. \tag{51}$$

Substituting (50) into (51): $r^K = 1$ and $w = 0$. No arbitrage between capital and loans, Kurlat (9.3.11):

$$1 + r = r^K + 1 - \delta. \tag{52}$$

Substitute $r^K = 1$ and $\delta = 1$ from (50) into (52):

$$1 + r = (1) + 1 - (1) = 1, \qquad r = 0.$$

Both prices are fixed by the linear technology, whatever K is.

Step — (c). The household's Euler equation:

$$u'(c_t) = \beta(1 + r)u'(c_{t+1}). \tag{53}$$

Substitute $r = 0$ from (b) into (53):

$$\begin{aligned} u'(c_t) &= \beta(1 + (0))u'(c_{t+1}) \\ &= \beta u'(c_{t+1}). \end{aligned}$$

As $\beta < 1$, $u'(c_t) < u'(c_{t+1})$ and, u' being decreasing, $c_{t+1} < c_t$: consumption falls over time. A steady state would need

$$F_K - \delta = \rho > 0 \tag{54}$$

(Kurlat (9.3.17)), but by (50) $F_K - \delta = 0$ always, so (54) never holds.

Economic intuition. The return on waiting ($r = 0$) is below the rate of time preference ($\rho > 0$), so the household runs the warehouse down.

Ref: Kurlat (2020), §9.1, eqs. (9.1.9)–(9.1.11); §9.3, eqs. (9.3.11), (9.3.17); project solution `Resolucao/kurlat_solutions_ch09.tex`, Problem 9.2.

Grading rubric.

- (b) w, r^K and r via no arbitrage: 3 pts; $r = 1$ (forgetting δ): max. 1 pt for (b).
- (c) Euler and sign: 2 pts. (a): 1 pt.

Question 6. (12 points) *Source: Kurlat, P. (2020),* A Course in Modern Macroeconomics*, Exercises 9.11 "Patience and Investment", p. 185, and 9.12 "Optimal vs Fixed Savings Rates", pp. 185–186 (the key synthesis) — statements reproduced verbatim, all items. Change: 9.12 is worked in Python (the book points to Matlab code); item (f) is added by this list. What this trains:* the saving rate Solow assumes is the one the optimising economy chooses in the long run; it is below s_{GR}.
Added (not in the book): work 9.11 symbolically, and 9.12 in Python (the book points to Matlab code).

9.11 Patience and Investment

Suppose the production function is $F(K, L) = K^\alpha L^{1-\alpha}$, and the labor supply is exogenous and equal to $L = 1$. The representative household's preferences are given by:

$$\sum_{t=0}^{\infty} \beta^t \frac{c_t^{1-\sigma}}{1-\sigma}$$

and the depreciation rate is δ.

(a) Use equation (9.3.17) to find an expression for the level of capital that this economy will have in the steady state.

(b) Find an expression for the level of GDP that the economy will have in the steady state.

(c) Find an expression for the steady-state investment-to-GDP ratio. How does it depend on β? Explain.

9.12 Optimal vs Fixed Savings Rates

Suppose the economy is as in Exercise 9.11, with $\alpha = 0.4$, $\beta = 0.95$, $\delta = 0.08$ and $\sigma = 2$. The economy starts with an initial capital stock $K_0 = 2$. We are going to compare how this economy behaves under a fixed investment rate and under an optimally-chosen investment rate.

(a) First suppose that, like we assumed in Chapter 4, the investment rate is exogenous and constant, and equal to the steady-state investment-to-GDP ratio that you found in Exercise 9.11. Compute GDP, consumption, investment and the capital stock for the first 150 years of this economy.

(b) Now suppose that consumption and investment are chosen in accordance with (9.3.13). Find the level of initial consumption that is consistent with the system of equations (9.3.15) - (9.3.13) and the economy converging to a steady state. (You can find a Matlab code that will compute this at the book website). Compute GDP, consumption, investment and the capital stock for the first 150 years of this economy.

(c) Plot the consumption paths of the two economies over the first 150 years on the same graph.

(d) Replace the computed paths of consumption from parts (a) and (b) into the utility function to compute how much utility the representative household will obtain in the first 150 years of each economy.

(e) By what fraction would one have to reduce consumption in the optimal-savings economy for the household to get the same utility as in the exogenous-savings economy?

Added parts (not in the book):

(f) Compare the steady-state investment rate with the golden-rule rate and the steady-state capital with K_{GR}.

Notation.

- K_t: capital (labour is fixed at 1), $Y_t = K_t^\alpha$: output, $\alpha \in (0,1)$. I_t: investment. $c_t > 0$: consumption.
- $\delta \in (0,1)$: depreciation. $\beta \in (0,1)$: discount factor; $\rho \equiv 1/\beta - 1 > 0$. $\sigma > 0$: CRRA curvature, $u(c) = \frac{c^{1-\sigma}}{1-\sigma}$. r_t: interest rate.
- K_{SS}, Y_{SS}, c_{SS}: steady-state values. $s \equiv I/Y$: investment (saving) rate. K_{GR}, s_{GR}: golden-rule capital and saving rate.
- 9.12: $\alpha = 0.4$, $\beta = 0.95$, $\delta = 0.08$, $\sigma = 2$, $K_0 = 2$, $t = 0, \ldots, 149$. U: discounted utility of a path. $\lambda > 0$: the factor that scales a whole consumption path.

Step — 9.11(a). The steady-state condition, Kurlat (9.3.17), with $F_K = \alpha K^{\alpha-1}$:

$$\alpha K_{SS}^{\alpha-1} - \delta = \frac{1}{\beta} - 1 = \rho. \tag{55}$$

Add δ to (55) and divide by α:

$$K_{SS}^{\alpha-1} = \frac{\rho + \delta}{\alpha}.$$

Raise both sides to the power $\frac{1}{\alpha-1}$:

$$K_{SS} = \left(\frac{\alpha}{\rho+\delta}\right)^{\frac{1}{1-\alpha}}. \tag{56}$$

σ does not appear.

Step — 9.11(b). Raise (56) to the power α:

$$\begin{aligned} Y_{SS} = K_{SS}^\alpha &= \left[\left(\frac{\alpha}{\rho+\delta}\right)^{\frac{1}{1-\alpha}}\right]^\alpha \\ &= \left(\frac{\alpha}{\rho+\delta}\right)^{\frac{\alpha}{1-\alpha}}, \end{aligned} \tag{57}$$

increasing in β (higher β, lower ρ).

Step — 9.11(c). In the steady state investment only replaces depreciation:

$$I = \delta K_{SS}. \tag{58}$$

Divide (58) by $Y_{SS} = K_{SS}^\alpha$, then use (56):

$$\begin{aligned} \frac{I}{Y} &= \frac{(\delta K_{SS})}{(K_{SS}^\alpha)} \\ &= \delta K_{SS}^{1-\alpha} \\ &= \delta \left[\left(\frac{\alpha}{\rho+\delta}\right)^{\frac{1}{1-\alpha}}\right]^{1-\alpha} \\ &= \frac{\alpha\delta}{\rho+\delta}. \end{aligned} \tag{59}$$

Step — 9.12 numbers. With $\alpha = 0.4$, $\beta = 0.95$, $\delta = 0.08$:

$$\rho = 1/0.95 - 1 = 0.052632, \qquad \rho + \delta = 0.132632, \qquad \alpha/(\rho+\delta) = 3.015873,$$

$$K_{SS} = 3.015873^{1/0.6} = 6.2954 \qquad \text{by (56)},$$

$$Y_{SS} = 6.2954^{0.4} = 2.0874 \qquad \text{by (57)},$$

$$s \equiv I/Y = (0.4)(0.08)/0.132632 = 0.241270 \qquad \text{by (59)},$$
$$c_{SS} = Y_{SS} - \delta K_{SS} = 2.0874 - (0.08)(6.2954) = 1.5838 \qquad \text{by (58)}.$$

Step — 9.12(a). The fixed-rate economy, with $1-\delta = 0.92$:

$$Y_t = K_t^{0.4}, \quad I_t = sY_t, \quad c_t = (1-s)Y_t, \quad K_{t+1} = 0.92K_t + I_t. \tag{60}$$

Iterate (60) from $K_0 = 2$:

$$\begin{aligned} Y_0 &= 2^{0.4} = 1.3195, \\ c_0 &= 0.758730 \times 1.3195 = 1.0012, \\ I_0 &= 0.241270 \times 1.3195 = 0.3184, \\ K_1 &= 0.92 \times 2 + 0.3184 = 2.1584. \end{aligned}$$

Step — 9.12(b). The optimal economy obeys the resource constraint and the Euler equation. With CRRA utility the Euler equation is $c_{t+1} = c_t[\beta(1+r_{t+1})]^{1/\sigma}$, and by no arbitrage $1 + r_{t+1} = \alpha K_{t+1}^{\alpha-1} + 1 - \delta$. Substituting the no-arbitrage rate into the Euler equation, then the parameters:

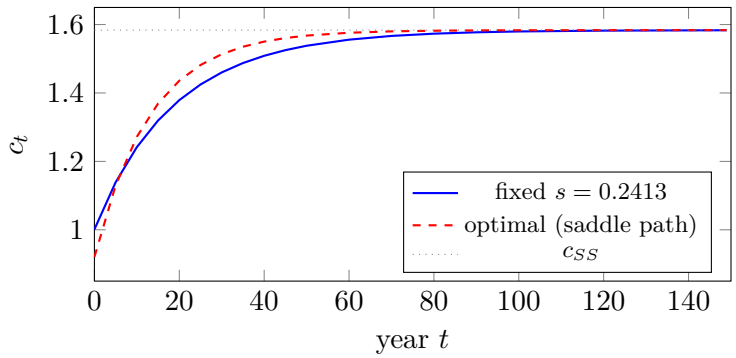
$$K_{t+1} = 0.92K_t + K_t^{0.4} - c_t, \tag{61}$$
$$\begin{aligned} c_{t+1} &= c_t\left[\beta\left(\alpha K_{t+1}^{\alpha-1} + 1 - \delta\right)\right]^{1/\sigma} \\ &= c_t\left[0.95\left(1 + 0.4K_{t+1}^{-0.6} - 0.08\right)\right]^{1/2}. \end{aligned} \tag{62}$$

Iterate (61)–(62) from a guess c_0; bisect: if K overshoots K_{SS}, raise c_0; if c starts falling (or K turns down) before K_{SS}, lower c_0. The saddle path gives $c_0 = 0.920627$. Then, by (61):

$$I_0 = 1.3195 - 0.9206 = 0.3989, \qquad K_1 = 0.92 \times 2 + 0.3989 = 2.2389.$$

Step — 9.12(c). Selected years (both economies start at $K_0 = 2$ and end at the same steady state):

t	K fixed	K optimal	c fixed	c optimal	I fixed	I optimal
0	2.0000	2.0000	1.0012	0.9206	0.3184	0.3989
5	2.7603	3.1169	1.1389	1.1297	0.3622	0.4461
10	3.4262	4.0048	1.2417	1.2719	0.3948	0.4701
25	4.8324	5.4958	1.4248	1.4818	0.4531	0.4953
50	5.8509	6.1667	1.5381	1.5678	0.4891	0.5025
149	6.2919	6.2954	1.5834	1.5838	0.5035	0.5036



The optimal economy consumes less at first and more later; the paths cross between years 5 and 6.

Step — 9.12(d). With $\sigma = 2$, $u(c) = \frac{c^{-1}}{-1} = -1/c$. Discounted utility of a path:

$$U = \sum_{t=0}^{149} 0.95^t u(c_t). \tag{63}$$

Evaluating (63) on the two paths of (c):

$$U^{\text{fixed}} = -15.79988, \qquad U^{\text{opt}} = -15.74360;$$

the optimal path is better.

Step — 9.12(e). Scaling every c_t by λ multiplies each $u(c_t)$ in (63), and so U, by $\lambda^{1-\sigma}$:

$$U(\lambda c) = \lambda^{1-\sigma} U(c) = \lambda^{-1} U(c). \tag{64}$$

Using (64), solve $\lambda^{-1} U^{\text{opt}} = U^{\text{fixed}}$:

$$\lambda = \frac{U^{\text{opt}}}{U^{\text{fixed}}} = \frac{(-15.74360)}{(-15.79988)} = 0.996438,$$

a cut of $1 - \lambda = 0.356\%$ of consumption.

Step — (f) golden rule. $s = 0.2413 < s_{GR} = \alpha = 0.40$. Rewrite (59):

$$\frac{I}{Y} = \alpha \cdot \frac{\delta}{\rho + \delta}, \qquad \frac{\delta}{\rho + \delta} < 1 \iff \rho > 0.$$

K_{GR} maximises steady-state consumption $K^\alpha - \delta K$, so it solves

$$F_K = \alpha K_{GR}^{\alpha - 1} = \delta. \tag{65}$$

Solve (65) as in 9.11(a):

$$K_{GR} = (\alpha/\delta)^{\frac{1}{1-\alpha}} = \big((0.4)/(0.08)\big)^{\frac{1}{1-(0.4)}} = 5^{1/0.6} = 14.62 > K_{SS} = 6.30.$$

Comparing (65) with (55), the gap closes only as $\beta \to 1$ ($\rho \to 0$).

Economic intuition. The right constant saving rate costs only a third of a percent of consumption: Solow's assumption is cheap when s is the equilibrium's long-run rate. That rate is set by impatience, not by the golden rule.

Ref: Kurlat (2020), §9.3, eqs. (9.3.13)–(9.3.17), pp. 172–178; §4.3, eq. (4.3.4). Numbers match the project's solution `Resolucao/kurlat_solutions_ch09.tex`, Problems 9.11–9.12 ($c_0 = 0.9206271904$, $\lambda = 0.996438$). The instructor's PSET 5 covers the Euler equation and arbitrage of the same economy (Lista 5, Q2) but stops before the steady state.

Grading rubric.

- 9.11 closed forms: 3 pts. 9.12(a)–(b) with a correct shooting rule: 4 pts; c_0 guessed rather than found: max. 1 pt for (b).
- (c)–(e): 3 pts. (f): 2 pts. Ceiling: claiming $K_{SS} = K_{GR}$ "because of the First Welfare Theorem": 0 for (f).

Question 7. (9 points) *Source: Romer, D. (2012),* Advanced Macroeconomics*, 4th ed., Problem 2.16 "Depreciation in the Diamond model and microeconomic foundations for the Solow model", p. 97 — statement reproduced verbatim, all items (a)–(b), preceded by the set-up and part (a) of Problem 2.15, p. 97, which 2.16(b) refers to (2.15(b)–(d) omitted); item (c) is added by this list. Harder, contrast. What this trains:* the exam's "Solow with $\delta = 1$" remark done right, in the overlapping-generations reading where the young save only their wage.

Added (not in the book): Problem 2.15 is included because 2.16(b) refers to it.

2.15. *A discrete-time version of the Solow model.* Suppose $Y_t = F(K_t, A_t L_t)$, with $F(\bullet)$ having constant returns to scale and the intensive form of the production function satisfying the Inada conditions. Suppose also that $A_{t+1} = (1+g)A_t$, $L_{t+1} = (1+n)L_t$, and $K_{t+1} = K_t + sY_t - \delta K_t$.

(a) Find an expression for k_{t+1} as a function of k_t.

2.16. *Depreciation in the Diamond model and microeconomic foundations for the Solow model.* Suppose that in the Diamond model capital depreciates at rate δ, so that $r_t = f'(k_t) - \delta$.

(a) How, if at all, does this change in the model affect equation (2.59) giving k_{t+1} as a function of k_t?

(b) In the special case of logarithmic utility, Cobb–Douglas production, and $\delta = 1$, what is the equation for k_{t+1} as a function of k_t? Compare this with the analogous expression for the discrete-time version of the Solow model with $\delta = 1$ from part (a) of Problem 2.15.

Added parts (not in the book):

(c) Compare the implied saving rate with s_{GR} and state when this economy over-accumulates.

Notation.

- C_{1t}, C_{2t+1}: consumption of generation t when young and when old. $\rho > -1$: discount rate, $\beta \equiv \frac{1}{1+\rho}$.
- A_t, L_t: technology and number of young, growing at rates g and n. $k \equiv K/(AL)$: capital per effective worker; $f(k)$: intensive production function, $f' > 0$, $f'' < 0$; Cobb–Douglas $f(k) = k^\alpha$, $\alpha \in (0, 1)$.
- w_t: wage per effective worker. r_{t+1}: interest rate. $\delta \in [0, 1]$: depreciation.
- $s(r)$: share of the wage the young save. s_S: Solow's saving rate that reproduces the same law of motion. s_{GR}: golden-rule rate.

Step — (a). Young at t maximise $\ln C_{1t} + \beta \ln C_{2t+1}$ subject to

$$C_{1t} + \frac{C_{2t+1}}{1 + r_{t+1}} = A_t w_t. \tag{66}$$

(66) does not contain δ, so the saving rule $s(r_{t+1})$ out of the wage is unchanged by depreciation. Only the return changes:

$$r_{t+1} = f'(k_{t+1}) - \delta. \tag{67}$$

Market clearing: next period's capital is the young's saving,

$$K_{t+1} = s(r_{t+1}) A_t w_t L_t. \tag{68}$$

The competitive wage per effective worker is

$$w = f(k) - k f'(k). \tag{69}$$

Divide (68) by $A_{t+1}L_{t+1} = (1+n)(1+g)A_t L_t$, then substitute (67) and (69):

$$\begin{aligned} k_{t+1} &= \frac{s(r_{t+1}) A_t w_t L_t}{((1+n)(1+g) A_t L_t)} \\ &= \frac{s(r_{t+1})\, w_t}{(1+n)(1+g)} \\ &= \frac{s(f'(k_{t+1}) - \delta)\,[f(k_t) - k_t f'(k_t)]}{(1+n)(1+g)}. \end{aligned} \tag{70}$$

In (70) depreciation matters only through $s(\cdot)$: in the log case not at all.

Step — (b). Log utility: the young save the share $\frac{\beta}{1+\beta}$ of the wage. Substitute $\beta = \frac{1}{1+\rho}$:

$$\begin{aligned} s &= \frac{(1/(1+\rho))}{1 + (1/(1+\rho))} \\ &= \frac{1/(1+\rho)}{(2+\rho)/(1+\rho)} \\ &= \frac{1}{2+\rho}, \end{aligned} \tag{71}$$

independent of r. Cobb–Douglas in (69):

$$\begin{aligned} w &= (k^\alpha) - k(\alpha k^{\alpha-1}) \\ &= k^\alpha - \alpha k^\alpha = (1-\alpha) k^\alpha. \end{aligned} \tag{72}$$

Substitute (71) and (72) into (70):

$$\begin{aligned} k_{t+1} &= \frac{\left(\frac{1}{2+\rho}\right)\left((1-\alpha)k_t^{\alpha}\right)}{(1+n)(1+g)} \\ &= \frac{(1-\alpha)k_t^{\alpha}}{(2+\rho)(1+n)(1+g)}. \end{aligned} \tag{73}$$

Solow 2.15(a) with $\delta = 1$:

$$k_{t+1} = \frac{s_S k_t^{\alpha}}{(1+n)(1+g)}. \tag{74}$$

(73) and (74) are identical with

$$\begin{aligned} s_S &= \frac{1-\alpha}{2+\rho} \\ &= \frac{1-\alpha}{2+\left(\frac{1}{\beta}-1\right)} \\ &= \frac{(1-\alpha)\beta}{1+\beta} : \end{aligned} \tag{75}$$

the young's saving of their wage is the whole gross saving, since with $\delta = 1$ the old hold no capital to run down.

Step — (c). Golden rule: $s_{GR} = \alpha$ (Question 2, (24)). Over-accumulation means (75) exceeds it. Multiply by $1+\beta$ and collect terms:

$$\begin{aligned} \frac{(1-\alpha)\beta}{1+\beta} &> \alpha \\ \beta - \alpha\beta &> \alpha + \alpha\beta \\ \beta(1-2\alpha) &> \alpha. \end{aligned}$$

As $\beta < 1$, the left side is below $1-2\alpha$, so this needs $1-2\alpha > \alpha$, i.e. $\alpha < \frac{1}{3}$. E.g. $\alpha = 0.2$, $\beta = 0.9$:

$$s_S = \frac{(1-0.2)(0.9)}{1+(0.9)} = 0.8 \times 0.9/1.9 = 0.379 > 0.2.$$

Unlike Kurlat's infinite-horizon economy (Q6(f)), finite lives can put the equilibrium above the golden rule.

Economic intuition. Diamond microfounds Solow's constant s only in a knife-edge case (log, Cobb–Douglas, $\delta = 1$), and the saving rate it delivers comes from preferences and the wage share, not from any golden-rule calculation.

Ref: Romer (2012), §§2.9–2.11, eq. (2.60), pp. 78–91; Romer, *Solution Manual*, Problems 2.15 and 2.16: same law of motion, $s(r) = 1/(2+\rho)$ and aggregate saving rate $(1-\alpha)/(2+\rho)$. Kurlat (2020), §9.1 (two-period equilibrium).

Grading rubric.

- (a) market clearing K_{t+1} = young's saving and the role of δ through r only: 3 pts.
- (b) law of motion and the identification $s_S = (1-\alpha)\beta/(1+\beta)$: 4 pts; $s_S = \beta/(1+\beta)$ (forgetting that only wages are saved): max. 2 pts.
- (c): 2 pts.

Question 8. (10 points) *Conceptual, words only.* True or false? Justify each in at most four lines, without algebra.

a) (2 points) In the exam's economy (log utility over c_1, c_2; one unit of time; work carries no disutility; $w_1 > 0$), no first-order condition for L_1 exists, so L_1 is left undetermined.

Answer: F

Notation.

- $L_1 \in [0,1]$: hours worked in period 1. $w_1 > 0$: wage. $\mathcal{L}$: the household's Lagrangian.
- $\lambda > 0$: multiplier on the budget (utility value of one unit of resources). $\nu \geq 0$: multiplier on $L_1 \leq 1$.

Step — why. L_1 is not in utility but it is in the budget: an extra hour adds w_1 to resources, worth $\lambda w_1 > 0$ in utility. The marginal gain is positive at every $L_1 < 1$, so the constraint $L_1 \leq 1$ binds: $L_1 = 1$, a corner. Stating "$\partial\mathcal{L}/\partial L_1 = \lambda w_1 - \nu = 0$ with $\nu > 0$, hence $L_1 = 1$" *is* the FOC. With leisure in utility it becomes interior (Q9).

b) (2 points) With $u = \ln c$ the saving rate does not depend on r; so in Kurlat 6.2 a rate cut leaves every person's $s = (y - c)/y$ unchanged.
Answer: F

Notation.
- $y_1, y_2 \geq 0$: income in periods 1 and 2; c_1: period-1 consumption; $s = (y - c)/y$: saving out of current income.
- $r > -1$: interest rate. $\beta \in (0,1)$: discount factor.

Step — why. Log makes saving a constant share $\frac{\beta}{1+\beta}$ of *lifetime wealth*, and c_1 independent of r only when all resources arrive in period 1 (the exam). With future income, wealth $y_1 + \frac{y_2}{1+r}$ rises when r falls, so c_1 rises and s out of current income falls: strongly for residents, hardly for athletes.

c) (2 points) The First Welfare Theorem holds in Kurlat's infinite-horizon economy, so its steady state is the golden rule.
Answer: F

Notation.
- F_K: marginal product of capital. $\delta \in (0,1)$: depreciation. $\rho > 0$: rate of time preference.
- K_{SS}: steady-state capital ($F_K = \rho + \delta$); K_{GR}: golden-rule capital ($F_K = \delta$).

Step — why. The golden rule maximises consumption *in the steady state*, ignoring the cost of getting there. Reaching it means giving up consumption now for gains later, which impatient households value less. Efficiency is relative to their preferences: the modified golden rule $F_K = \rho + \delta > \delta$ puts K_{SS} below K_{GR}. No contradiction.

d) (2 points) An economy with capital above the golden rule is dynamically inefficient: a permanently lower saving rate raises consumption at every date, starting today.
Answer: T

Notation.
- s: saving rate. k: capital per (effective) worker; k_{GR}: its golden-rule level.
- r: interest rate. n, g: growth rates of population and technology.

Step — why. Cutting s raises consumption at once; as capital falls toward k_{GR}, steady-state consumption rises (it falls in k above k_{GR}), so no date loses. Equivalently $r < n$ $(+g)$. This Pareto improvement cannot exist in Kurlat's economy (Q6(f)) but can in overlapping generations (Q7(c), Q10), where infinitely many generations break the welfare theorem.

e) (2 points) Once the household's rule $a_1 = \frac{\beta}{1+\beta} \times$(resources) is known, the economy's law of motion $k_2 = \phi(k_1)$ follows without further conditions.
Answer: F

Notation.
- a_1: the household's saving (assets carried to period 2). k_1, k_2 (K_1, K_2): capital in periods 1 and 2; ϕ: the law of motion $k_2 = \phi(k_1)$.
- w_1: wage. Π_1: profit paid to the household that owns the firm. Y_1: period-1 output.

Step — why. a_1 is a demand for assets at given prices. Three more pieces are needed: firm FOCs that turn k_1 into w_1 and Π_1 (resources $= Y_1$ when the household owns the firm), market clearing $K_2 = a_1$ (capital is the only asset), and the period timing. Only then is saving a law of motion.

Grading rubric.
- Each item: 0.5 for the verdict, 1.5 for the reason. A right verdict with a wrong reason: 0.5.

Block 3 — Created: the exam's economy, extended

(Kurlat §§6.2, 7.1, 9.1, 4.3; Romer §2.9–2.11 • 28 points)

Question 9. (15 points) *The exam's household with leisure, inside an economy.* Periods $t = 1, 2, \ldots$ Each period a generation of size one is born and lives two periods, with one unit of time when young and none when old. Generation t maximises $\ln c_{1t} + \psi \ln(1 - L_t) + \beta \ln c_{2t+1}$, $\psi > 0$, $0 < \beta < 1$. Technology $Y_t = AK_t^{\alpha} L_t^{1-\alpha}$, $\delta = 1$, competitive markets; saving is lent to firms as next period's capital and earns $1 + r_{t+1} = r_{t+1}^K$. *Generation 1 is the exam's household:* it is born with assets $a_0 \geq 0$ and owns the firm, which owns $K_1 > 0$, so in period 1 it receives the profit $\Pi_1 = Y_1 - w_1 L_1$ besides its wage. From period 2 on, the old rent their capital to firms, which earn zero profit; each young generation earns only its wage.

a) (2 points) Write generation 1's two period budgets and its present-value budget constraint. List every choice variable and every income source, and read the constraint in words.

b) (4 points) Derive all first-order conditions and closed forms for c_1, c_2, L_1, a_1 given (w_1, Π_1, r_2, a_0). When is L_1 interior? Show that a_1 is the share $\frac{\beta}{1+\beta}$ of the resources $a_0 + w_1 L_1 + \Pi_1$. What happens as $\psi \to 0$?

c) (3 points) Let $a_0 = 0$. Using the firm's FOC, find equilibrium L_1^* (show it depends on neither K_1 nor A) and the map $K_2 = \phi_1(K_1)$.

d) (3 points) For $t \geq 2$ find the young's labour $\bar{L}$, the law of motion $K_{t+1} = \phi(K_t)$, the steady state and its stability. Draw ϕ against the 45° line for $\alpha = 1/3$, $\beta = 0.8$, $\psi = 0.5$, $A = 5$, $K_1 = 1$, marking K^* and the path from K_2.

e) (3 points) Compare with Solow (δ, n, s) and with s_{GR}. When does this economy over-accumulate? Evaluate for $(\alpha, \beta) = (1/3, 0.8)$ and $(0.2, 0.9)$, with the interest rate in each. Why does generation 1 work less than later ones, and what goes wrong if Π_1 is left out of its budget?

Notation.

- c_{1t}, $c_{2t+1} > 0$: consumption of generation t young and old (for generation 1: c_1, c_2). $L_t \in [0, 1]$: hours of the young; $1 - L_t$: leisure. $\psi > 0$: weight of leisure. $\beta \in (0, 1)$: discount factor.
- $Y_t = AK_t^{\alpha} L_t^{1-\alpha}$: output, $A > 0$, $\alpha \in (0, 1)$; K_t: capital, $\delta = 1$.
- w_t: wage. r_{t+1}: interest rate, $1 + r_{t+1} = r_{t+1}^K$ (rental rate). $a_0 \geq 0$: generation 1's initial assets. a_1 (a_t): saving of generation 1 (t). $\Pi_1 = Y_1 - w_1 L_1$: period-1 profit.
- X: full wealth; R: resources; $\lambda > 0$: multiplier on the budget (defined where first used).
- L_1^*: generation 1's equilibrium hours; $\bar{L}$: later generations' hours. ϕ_1, ϕ: laws of motion of capital; K^*: steady state. s_{eq}: the equilibrium saving rate K_{t+1}/Y_t. s_{GR}: golden-rule rate.

Step — (a). The two period budgets of generation 1:

$$c_1 + a_1 = a_0 + w_1 L_1 + \Pi_1, \tag{76}$$

$$c_2 = (1 + r_2) a_1. \tag{77}$$

Solve (77) for $a_1 = \frac{c_2}{1+r_2}$ and substitute into (76):

$$c_1 + \frac{c_2}{1 + r_2} = a_0 + w_1 L_1 + \Pi_1. \tag{78}$$

Add $w_1(1 - L_1)$ to both sides of (78):

$$c_1 + \frac{c_2}{1 + r_2} + w_1(1 - L_1) = a_0 + w_1 + \Pi_1 \equiv X. \tag{79}$$

Choices: c_1, c_2, a_1, L_1 with $0 \leq L_1 \leq 1$. Income: initial wealth, labour, *and profits*. Reading: the present value of consumption plus the value of leisure equals full wealth (time valued at w_1).

Step — (b) FOCs. The Lagrangian, with the budget (78):

$$\mathcal{L} = \ln c_1 + \psi \ln(1 - L_1) + \beta \ln c_2 + \lambda \Big[a_0 + w_1 L_1 + \Pi_1 - c_1 - \frac{c_2}{1 + r_2} \Big]. \tag{80}$$

Differentiate (80) with respect to c_1, c_2 and L_1:

$$\frac{1}{c_1} = \lambda, \tag{81}$$

$$\frac{\beta}{c_2} = \frac{\lambda}{1+r_2}, \tag{82}$$

$$\frac{\psi}{1-L_1} = \lambda w_1. \tag{83}$$

Substitute (81) into (82) (Euler) and into (83) (labour supply):

$$\frac{\beta}{c_2} = \frac{(1/c_1)}{1+r_2}$$

$$c_2 = \beta(1+r_2)c_1, \tag{84}$$

$$\frac{\psi}{1-L_1} = \Big(\frac{1}{c_1}\Big)w_1$$

$$w_1(1-L_1) = \psi c_1. \tag{85}$$

Step — (b) closed forms. Substitute (84) and (85) into the full budget (79):

$$c_1 + \frac{(\beta(1+r_2)c_1)}{1+r_2} + (\psi c_1) = X$$

$$c_1 + \beta c_1 + \psi c_1 = X,$$

so

$$c_1 = \frac{X}{1+\beta+\psi}. \tag{86}$$

Substitute (86) into (84), into (85) solved for L_1, and into $a_1 = \frac{c_2}{1+r_2}$:

$$c_2 = \beta(1+r_2)\Big(\frac{X}{1+\beta+\psi}\Big)$$

$$= \frac{\beta(1+r_2)X}{1+\beta+\psi},$$

$$L_1 = 1 - \frac{\psi}{w_1}\Big(\frac{X}{1+\beta+\psi}\Big)$$

$$= 1 - \frac{\psi X}{(1+\beta+\psi)w_1}, \tag{87}$$

$$a_1 = \frac{1}{1+r_2}\Big(\frac{\beta(1+r_2)X}{1+\beta+\psi}\Big)$$

$$= \frac{\beta X}{1+\beta+\psi}. \tag{88}$$

Interior iff $L_1 > 0$. By (87) and the definition of X in (79):

$$L_1 > 0 \iff 1 - \frac{\psi(a_0+w_1+\Pi_1)}{(1+\beta+\psi)w_1} > 0$$

$$\iff \psi(a_0+w_1+\Pi_1) < (1+\beta+\psi)w_1$$

$$\iff \psi(a_0+\Pi_1) < (1+\beta)w_1. \tag{89}$$

Step — (b) saving share. Resources are full wealth minus the value of leisure, (79); then use (85) and (86):

$$R \equiv a_0 + w_1 L_1 + \Pi_1 = X - w_1(1-L_1)$$

$$
\begin{aligned}
&= X - (\psi c_1) \\
&= X - \psi\Big(\frac{X}{1+\beta+\psi}\Big) \\
&= X - \frac{\psi X}{1+\beta+\psi} \\
&= \frac{(1+\beta)X}{1+\beta+\psi}.
\end{aligned} \tag{90}
$$

Divide (88) by (90):

$$
\begin{aligned}
\frac{a_1}{R} &= \frac{\big(\beta X/(1+\beta+\psi)\big)}{\big((1+\beta)X/(1+\beta+\psi)\big)} \\
&= \frac{\beta}{1+\beta},
\end{aligned}
$$

the exam's rate, for any ψ and any r_2. As $\psi \to 0$, (87) and (86) give

$$
\begin{aligned}
L_1 &\to 1 - \frac{(0)X}{(1+\beta+(0))w_1} = 1 \quad \text{(the exam's corner)}, \\
c_1 &\to \frac{(a_0 + w_1 + \Pi_1)}{1+\beta+(0)} = \frac{a_0 + w_1 + \Pi_1}{1+\beta}.
\end{aligned}
$$

Step — (c) labour. The firm's FOC for labour, and profit as output minus wages:

$$
w_1 = (1-\alpha)AK_1^{\alpha}L_1^{-\alpha} = \frac{(1-\alpha)Y_1}{L_1}, \tag{91}
$$

$$
\Pi_1 = Y_1 - w_1 L_1 = \alpha Y_1. \tag{92}
$$

With $a_0 = 0$, (79) and (92) give $X = (0) + w_1 + (\alpha Y_1) = w_1 + \alpha Y_1$. Multiply (87) out as $w_1(1-L_1) = \frac{\psi X}{1+\beta+\psi}$ and substitute (91) and X:

$$
\begin{aligned}
\Big(\frac{(1-\alpha)Y_1}{L_1}\Big)(1-L_1) &= \frac{\psi}{1+\beta+\psi}\Big[\Big(\frac{(1-\alpha)Y_1}{L_1}\Big) + \alpha Y_1\Big] \\
\frac{(1-\alpha)Y_1(1-L_1)}{L_1} &= \frac{\psi}{1+\beta+\psi}\Big[\frac{(1-\alpha)Y_1}{L_1} + \alpha Y_1\Big].
\end{aligned}
$$

Multiply by $\frac{L_1(1+\beta+\psi)}{Y_1}$:

$$
(1-\alpha)(1+\beta+\psi)(1-L_1) = \psi(1-\alpha) + \psi\alpha L_1.
$$

Collect the L_1 terms on the right; $\psi(1-\alpha)$ cancels on the left:

$$
(1-\alpha)(1+\beta) = L_1\big[(1-\alpha)(1+\beta) + \psi\big].
$$

Hence

$$
L_1^{*} = \frac{(1-\alpha)(1+\beta)}{(1-\alpha)(1+\beta)+\psi}. \tag{93}
$$

Y_1 cancelled, so K_1 and A drop out. Interiority: (89) with $a_0 = 0$, then (91)–(92), then (93):

$$
\begin{aligned}
\psi\Pi_1 < (1+\beta)w_1 &\iff \psi(\alpha Y_1) < (1+\beta)\Big(\frac{(1-\alpha)Y_1}{L_1^{*}}\Big) \\
&\iff \psi\alpha L_1^{*} < (1-\alpha)(1+\beta) \\
&\iff \psi\alpha\Big(\frac{(1-\alpha)(1+\beta)}{(1-\alpha)(1+\beta)+\psi}\Big) < (1-\alpha)(1+\beta)
\end{aligned}
$$

$$\iff \psi\alpha < (1-\alpha)(1+\beta)+\psi,$$

always true.

Step — (c) map. With $a_0 = 0$, (91)–(92) give $R = w_1L_1 + \Pi_1 = (\frac{(1-\alpha)Y_1}{L_1})L_1 + (\alpha Y_1) = Y_1$. Capital is the only asset, so market clearing is

$$K_2 = a_1. \tag{94}$$

Substitute $a_1 = \frac{\beta}{1+\beta}R = \frac{\beta}{1+\beta}Y_1$ and $L_1 = L_1^*$ into (94):

$$\begin{aligned} K_2 &= \Big(\frac{\beta}{1+\beta}Y_1\Big) \\ &= \frac{\beta}{1+\beta}(AK_1^{\alpha}(L_1^*)^{1-\alpha}) \\ &= \frac{\beta}{1+\beta}AK_1^{\alpha}(L_1^*)^{1-\alpha} \equiv \phi_1(K_1). \end{aligned} \tag{95}$$

Step — (d) later generations. $a_0 = \Pi = 0$, so (79) gives $X = (0) + w_t + (0) = w_t$, and (87) becomes

$$\begin{aligned} \bar{L} &= 1 - \frac{\psi(w_t)}{(1+\beta+\psi)w_t} \\ &= 1 - \frac{\psi}{1+\beta+\psi} = \frac{1+\beta}{1+\beta+\psi}. \end{aligned} \tag{96}$$

Saving is the share $\frac{\beta}{1+\beta}$ of resources $w_t\bar{L}$, and $w_t\bar{L} = (1-\alpha)Y_t$ by (91):

$$a_t = \frac{\beta}{1+\beta}w_t\bar{L} = \frac{\beta}{1+\beta}((1-\alpha)Y_t). \tag{97}$$

Market clearing $K_{t+1} = a_t$, with $Y_t = AK_t^{\alpha}\bar{L}^{1-\alpha}$ in (97):

$$\begin{aligned} K_{t+1} &= \frac{\beta}{1+\beta}(1-\alpha)(AK_t^{\alpha}\bar{L}^{1-\alpha}) \\ \phi(K) &= \frac{\beta(1-\alpha)}{1+\beta}A\bar{L}^{1-\alpha}K^{\alpha}. \end{aligned} \tag{98}$$

$\phi(0) = 0$, $\phi' > 0$, $\phi'' < 0$, $\phi'(0^+) = \infty$, $\phi'(\infty) = 0$: one positive fixed point. Set $\phi(K) = K$ in (98), divide by K^{α} and raise to the power $\frac{1}{1-\alpha}$:

$$\begin{aligned} \frac{\beta(1-\alpha)}{1+\beta}A\bar{L}^{1-\alpha}K^{\alpha} &= (K) \\ K^{1-\alpha} &= \frac{\beta(1-\alpha)A\bar{L}^{1-\alpha}}{1+\beta} \\ K^* &= \Big[\frac{\beta(1-\alpha)A\bar{L}^{1-\alpha}}{1+\beta}\Big]^{\frac{1}{1-\alpha}}. \end{aligned} \tag{99}$$

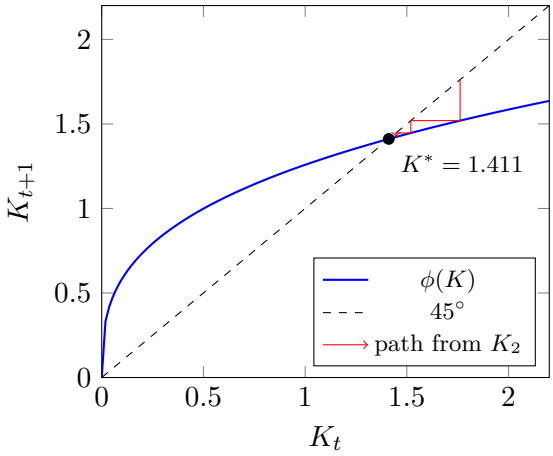
Differentiating (98), $\phi'(K^*) = \alpha\frac{\phi(K^*)}{K^*} = \alpha\frac{(K^*)}{K^*} = \alpha < 1$: monotone, stable.

Step — (d) numbers. With $\alpha = 1/3$, $\beta = 0.8$, $\psi = 0.5$, $A = 5$, $K_1 = 1$:

$$\begin{aligned} \bar{L} &= \tfrac{1+0.8}{1+0.8+0.5} = 1.8/2.3 = 0.7826 && \text{by (96)}, \\ L_1^* &= \tfrac{(2/3)(1.8)}{(2/3)(1.8)+0.5} = 1.2/1.7 = 0.7059 && \text{by (93)}, \\ \phi(K) &= \tfrac{0.8\cdot 2/3}{1.8}\cdot 5\cdot 0.7826^{2/3}K^{1/3} = 1.2581K^{1/3} && \text{by (98)}, \\ K^* &= 1.2581^{1.5} = 1.4112 && \text{by (99)}, \end{aligned}$$

$$Y_1 = 5 \cdot (1)^{1/3} \cdot 0.7059^{2/3} = 3.9639,$$
$$K_2 = \tfrac{0.8}{1.8} \times 3.9639 = 1.7617 \qquad \text{by (95)}.$$

Then (98) gives $K_3 = 1.2581\,(1.7617)^{1/3} = 1.5195$ and, iterating the same way, $K_4 = 1.4464$, $K_5 = 1.4228 \to K^*$.



Step — (e) Solow. With $\delta = 1$, $n = 0$ Solow's law of motion is

$$K_{t+1} = sY_t. \tag{100}$$

Here, by market clearing and (97),

$$\frac{K_{t+1}}{Y_t} = \frac{\beta(1-\alpha)}{1+\beta} \equiv s_{eq}: \tag{101}$$

the same law of motion as (100), but s is chosen (patience, and the wage share because only wages are saved), and it need not be $s_{GR} = \alpha$. Golden rule: MPK $= n + \delta = 1$. At K^*, $K = s_{eq}Y$ by (101), so

$$\begin{aligned} 1 + r = \text{MPK} &= \alpha \frac{Y}{K} \\ &= \alpha \frac{Y}{(s_{eq}Y)} \\ &= \frac{\alpha}{s_{eq}}. \end{aligned} \tag{102}$$

By (102), and multiplying out (101) as in Question 7(c):

$$\begin{aligned} r < 0 = n &\iff s_{eq} > \alpha \\ &\iff \frac{\beta(1-\alpha)}{1+\beta} > \alpha \\ &\iff \beta(1-2\alpha) > \alpha. \end{aligned}$$

Step — (e) numbers. Using (101) and (102):

$$(1/3, 0.8): \; s_{eq} = \tfrac{0.8\times 2/3}{1.8} = 0.2963 < 0.3333, \quad 1 + r = \tfrac{1/3}{0.2963} = 1.125, \quad r = 0.125 > 0,$$
$$(0.2, 0.9): \; s_{eq} = \tfrac{0.9\times 0.8}{1.9} = 0.3789 > 0.2, \quad 1 + r = \tfrac{0.2}{0.3789} = 0.5278, \quad r = -0.472 < 0.$$

The first is under-accumulation, efficient; the second is over-accumulation, dynamically inefficient.

Step — (e) profits. Compare (93) with (96); cross-multiply and cancel $(1-\alpha)(1+\beta)$:

$$L_1^* < \bar{L} \iff \frac{(1-\alpha)(1+\beta)}{(1-\alpha)(1+\beta)+\psi} < \frac{1+\beta}{1+\beta+\psi}$$

$$\iff (1-\alpha)(1+\beta+\psi) < (1-\alpha)(1+\beta)+\psi$$
$$\iff (1-\alpha)\psi < \psi,$$

true. Profits are non-labour income; the wealth effect buys leisure. Dropping Π_1 gives $L_1 = \bar{L}$ and $K_2 = \frac{\beta}{1+\beta}(1-\alpha)Y_1$: it understates resources by αY_1 and overstates labour.

Economic intuition. The household rule is the same in every generation; what differs is who owns what. Market clearing turns that rule into ϕ, and the saving rate it implies answers to preferences and factor shares, not to the golden rule.

Ref: Kurlat (2020), §6.2 (Euler), §7.1 (labour supply), §9.1 (two-period equilibrium), §4.3 (golden rule); Romer (2012), §§2.9–2.11. Same structure as Prova 01, Q2(c)–(f), with $\psi = 0$; see `Prova/prova-01-error-analysis.md`.

Grading rubric.

- (a): 2 pts; Π_1 missing: max. 1 pt. (b) all four FOCs (or three plus the corner argument): 2 pts; closed forms and $\beta/(1+\beta)$: 2 pts.
- (c): 3 pts. (d): 3 pts, of which the diagram 1 pt (axes, 45° line, curve, K^* and direction marked separately: 0.25 each).
- (e): 3 pts. Ceiling: no market clearing ($K_{t+1} = a_t$) anywhere: max. 6 pts for the whole question.

Question 10. (13 points) *CRRA and population growth: when the saving rate depends on r.* Overlapping generations as in Question 9 but without leisure: each young person supplies one unit of labour, earns only the wage, and the number of young grows, $L_{t+1} = (1+n)L_t$. Utility $u(c_{1t}) + \beta u(c_{2t+1})$ with $u(c) = \frac{c^{1-\sigma}}{1-\sigma}$; $y = Ak^\alpha$ per young worker; $\delta = 1$.

a) (3 points) Solve the young's problem. Show that saving is a share $s(r)$ of the wage, find $s(r)$, and sign $s'(r)$. Explain the sign through income and substitution effects.

b) (3 points) Using the firm's FOCs and market clearing $K_{t+1} = L_t a_t$, write the law of motion for $k \equiv K/L$. Show that k_{t+1} is unique given k_t when $\sigma \leq 1$, and give it in closed form for $\sigma = 1$.

c) (3 points) Show that the steady state satisfies $s(r)(1+r) = \frac{\alpha(1+n)}{1-\alpha}$ and that the aggregate saving rate is $\frac{I}{Y} = \frac{\alpha(1+n)}{1+r}$. Conclude: $\frac{I}{Y} > s_{GR} \iff r < n \iff k^* > k_{GR}$.

d) (2 points) Let $\alpha = 0.2$, $\beta = 0.5$, $n = 0.3$, $A = 5$. For $\sigma = 1$ find r^*, k^*, k_{GR} and steady-state consumption per worker at both. For $\sigma \in \{1/2, 2\}$ find r^* numerically (Python) and say which economies are dynamically inefficient, and why σ matters.

e) (2 points) Draw the $\sigma = 1$ map against the 45° line, marking k^* and k_{GR}. In what sense is this "Solow with $\delta = 1$"?

Notation.

- c_1, $c_2 > 0$: consumption young and old. w: wage. a: saving of a young person. $r > -1$: interest rate; r^K: rental rate.
- $u(c) = \frac{c^{1-\sigma}}{1-\sigma}$, $\sigma > 0$ ($1/\sigma$: intertemporal elasticity). $\beta \in (0,1)$: discount factor.
- D: the consumption divisor, as in Question 3, (42). $s(r)$: share of the wage saved.
- L_t: number of young, growing at rate $n > -1$. $k \equiv K/L$: capital per young worker; $y = Ak^\alpha$, $A > 0$, $\alpha \in (0,1)$; $\delta = 1$.
- I/Y: aggregate saving (investment) rate. k^*, r^*: steady state. k_{GR}, s_{GR}: golden rule. $H(r)$: the steady-state residual of (d). $\phi(k)$: the law of motion; s_S: Solow's saving rate that reproduces it.

Step — (a). The young person's two budgets:

$$c_1 + a = w, \tag{103}$$
$$c_2 = (1+r)a. \tag{104}$$

The Euler equation is (40), $c_2 = [\beta(1+r)]^{1/\sigma} c_1$. Eliminating a from (103)–(104) gives the PV budget

$$c_1 + \frac{c_2}{1+r} = w. \tag{105}$$

(105) is (41) with $W = w$, so, as in (43),

$$c_1 = \frac{w}{D}, \qquad D = 1 + \beta^{1/\sigma}(1+r)^{\frac{1}{\sigma}-1}. \tag{106}$$

Substitute (106) into (103):

$$\begin{aligned} \Big(\frac{w}{D}\Big) + a &= w \\ a &= w - \frac{w}{D} = s(r)w, \end{aligned} \tag{107}$$

with

$$\begin{aligned} s(r) &= \frac{D-1}{D} \\ &= \frac{(1+\beta^{1/\sigma}(1+r)^{\frac{1}{\sigma}-1}) - 1}{(1+\beta^{1/\sigma}(1+r)^{\frac{1}{\sigma}-1})} \\ &= \frac{\beta^{1/\sigma}(1+r)^{\frac{1}{\sigma}-1}}{1+\beta^{1/\sigma}(1+r)^{\frac{1}{\sigma}-1}}. \end{aligned} \tag{108}$$

Write (108) as $s = 1 - 1/D$ and differentiate, with $D'(r)$ from (45):

$$s'(r) = \frac{D'(r)}{D^2}, \qquad D'(r) = \beta^{1/\sigma}\Big(\frac{1}{\sigma} - 1\Big)(1+r)^{\frac{1}{\sigma}-2}. \tag{109}$$

The sign of (109) is the sign of $1 - \sigma$. The young are pure savers: a higher r makes them richer ($c_1 \uparrow$, saving $\downarrow$) and makes waiting cheaper (saving $\uparrow$). $\sigma < 1$: substitution wins; $\sigma > 1$: income wins; $\sigma = 1$: (108) gives

$$\begin{aligned} s &= \frac{\beta^{1/(1)}(1+r)^{\frac{1}{(1)}-1}}{1+\beta^{1/(1)}(1+r)^{\frac{1}{(1)}-1}} \\ &= \frac{\beta}{1+\beta}, \end{aligned}$$

independent of r.

Step — (b). The firm's FOCs, with $\delta = 1$ so that $1 + r = r^K$:

$$w_t = (1-\alpha)Ak_t^{\alpha}, \tag{110}$$

$$1 + r_{t+1} = r^K_{t+1} = \alpha A k_{t+1}^{\alpha-1}. \tag{111}$$

Market clearing:

$$K_{t+1} = L_t a_t. \tag{112}$$

Divide (112) by $L_{t+1} = (1+n)L_t$ and use (107):

$$(1+n)k_{t+1} = a_t = s(r_{t+1})w_t.$$

Substitute (110) and (111):

$$(1+n)k_{t+1} = s\big(\alpha A k_{t+1}^{\alpha-1} - 1\big)(1-\alpha)Ak_t^{\alpha}. \tag{113}$$

The left side of (113) rises from 0 to ∞ in k_{t+1}; when $\sigma \leq 1$ the right side is positive and non-increasing in k_{t+1} (higher k_{t+1}, lower r, s weakly lower by (109)): one crossing. $\sigma = 1$: $s = \frac{\beta}{1+\beta}$ in (113), divided by $1 + n$:

$$(1+n)k_{t+1} = \Big(\frac{\beta}{1+\beta}\Big)(1-\alpha)Ak_t^{\alpha}$$

$$k_{t+1} = \frac{\beta(1-\alpha)A}{(1+\beta)(1+n)} k_t^{\alpha}. \tag{114}$$

Step — (c). Steady state: set $k_{t+1} = k_t = k$ in (113) and divide by k^{α}:

$$\begin{aligned}(1+n)(k) &= s\big(\alpha A(k)^{\alpha-1} - 1\big)(1-\alpha)A(k)^{\alpha} \\ (1+n)k^{1-\alpha} &= s(r)(1-\alpha)A. \end{aligned} \tag{115}$$

Solve (111) for $k^{1-\alpha}$:

$$k^{1-\alpha} = \frac{\alpha A}{1+r}. \tag{116}$$

Substitute (116) into (115):

$$\begin{aligned}(1+n)\Big(\frac{\alpha A}{1+r}\Big) &= s(r)(1-\alpha)A \\ \frac{(1+n)\alpha A}{1+r} &= s(r)(1-\alpha)A \\ s(r)(1+r) &= \frac{\alpha(1+n)}{1-\alpha}. \end{aligned} \tag{117}$$

Gross investment is K_{t+1} ($\delta = 1$), which per young worker of period t is $(1+n)k$. Divide by $y = Ak^{\alpha}$, then use (116):

$$\begin{aligned}\frac{I}{Y} &= \frac{(1+n)k}{Ak^{\alpha}} \\ &= \frac{(1+n)k^{1-\alpha}}{A} \\ &= \frac{1+n}{A}\Big(\frac{\alpha A}{1+r}\Big) \\ &= \frac{\alpha(1+n)}{1+r}. \end{aligned} \tag{118}$$

By (117), (118) also equals

$$\frac{I}{Y} = \frac{\big((1-\alpha)s(r)(1+r)\big)}{1+r} = (1-\alpha)s(r).$$

Golden rule: MPK $= n + \delta = 1 + n$; by (111) that is $1 + r = 1 + n$, where (118) gives $\frac{I}{Y} = \frac{\alpha(1+n)}{(1+n)} = \alpha = s_{GR}$. As $\frac{I}{Y}$ falls in r and MPK falls in k: $\frac{I}{Y} > \alpha \iff r < n \iff k^* > k_{GR}$.

Step — (d) $\sigma = 1$. With $s = \frac{\beta}{1+\beta}$, solve (117) for $1 + r$:

$$\begin{aligned}\Big(\frac{\beta}{1+\beta}\Big)(1+r^*) &= \frac{\alpha(1+n)}{1-\alpha} \\ 1 + r^* &= \frac{\alpha(1+n)(1+\beta)}{(1-\alpha)\beta} = \frac{0.2 \times 1.3 \times 1.5}{0.8 \times 0.5} = 0.975. \end{aligned}$$

So $r^* = -0.025 < n = 0.3$: over-accumulation. By (118):

$$\frac{I}{Y} = \frac{(0.2)(1.3)}{(0.975)} = \frac{0.26}{0.975} = 0.2667 > 0.2.$$

The fixed point of (114), and (116) at $r = n$:

$$k^* = \Big(\frac{0.5 \times 0.8 \times 5}{1.5 \times 1.3}\Big)^{1.25} = 1.0256^{1.25} = 1.0322,$$

$$k_{GR} = \Big(\frac{\alpha A}{1+n}\Big)^{1.25} = \Big(\frac{(0.2)(5)}{1+(0.3)}\Big)^{1.25} = (1/1.3)^{1.25} = 0.7204.$$

Steady-state consumption per worker is output minus the investment that keeps k constant:

$$c = Ak^{\alpha} - (1+n)k. \tag{119}$$

Evaluating (119):

$$c(k^*) = 5(1.0322)^{0.2} - 1.3(1.0322) = 3.6899,$$
$$c(k_{GR}) = 5(0.7204)^{0.2} - 1.3(0.7204) = 3.7461,$$

so $c(k^*) < c(k_{GR})$.

Step — (d) other σ. Write (117) as a root problem:

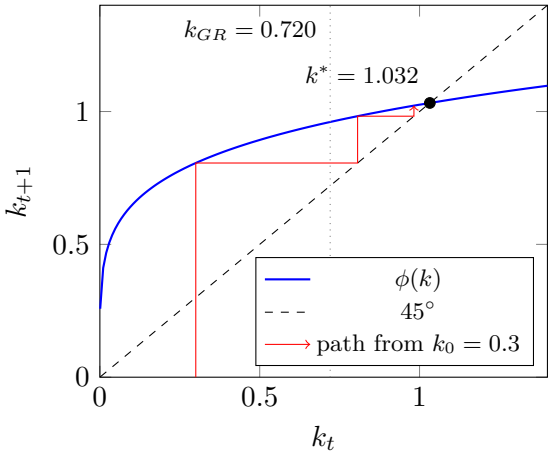
$$H(r) = (1-\alpha)s(r)(1+r) - \alpha(1+n) = 0. \tag{120}$$

Solve (120) by bisection on $(-0.99, 49)$ (one sign change on a grid), and evaluate (118):

$$\sigma = 1/2: \ r^* = 0.314 > n, \quad \tfrac{I}{Y} = 0.198 < 0.2 \text{ (efficient)},$$
$$\sigma = 2: \ r^* = -0.287 < n, \quad \tfrac{I}{Y} = 0.365 \text{ (inefficient)}.$$

With high σ the young save *more* when r is low, which pushes k up and r further down; with low σ low returns discourage saving and hold k back.

Step — (e). With the numbers of (d), (114) is $\phi(k) = 1.0256k^{0.2}$. It crosses the 45° line at $k^* = 1.032$, to the right of $k_{GR} = 0.720$; from $k_0 = 0.3$: 0.806, 0.982, 1.022, 1.030.



Solow's discrete law with $\delta = 1$ is

$$k_{t+1} = \frac{s_S A k_t^{\alpha}}{1+n}. \tag{121}$$

(114) is (121) with $s_S = \frac{(1-\alpha)\beta}{1+\beta}$; with $\sigma \neq 1$, (113) shows that the rate $s_S = (1-\alpha)s(r_{t+1})$ moves along the transition, so Solow's constant s holds only in the log case.

Economic intuition. $r < n$ says capital earns less than the economy's growth rate: a chain of transfers from young to old would beat saving. Whether the market gets there depends on how saving responds to r, which is σ.

Ref: Kurlat (2020), §6.2 and Exercise 6.1(e) (sign of $\partial c_1/\partial r$); §4.3 (golden rule with n). Romer (2012), §§2.9–2.10, eqs. (2.55)–(2.60), pp. 78–88, and §2.11 (dynamic inefficiency), pp. 88–91; Romer, *Solution Manual*, Problem 2.16(b) for the log case.

Grading rubric.

- (a) $s(r)$ and the sign of s' with its reading: 3 pts; sign read as $\sigma - 1$: max. 1 pt.
- (b): 3 pts; no market clearing: 0 for (b). (c): 3 pts; identity $I/Y = \alpha(1+n)/(1+r)$ quoted: max. 1 pt.
- (d): 2 pts. (e) diagram (axes, 45°, curve, k^*, k_{GR}) and the Solow reading: 2 pts.